SAFE TEST PLAN · CONTROL EDITION V1.4

Testing the D2 on the Cerbo, one safe step at a time

Each stage only adds a little risk, and only after the stage before it has passed. Stages 1–4 can't change what the D2 does at all, because the Cerbo sends it nothing. Stop at the first thing that looks wrong.

Before stage 5 on a D2: this driver was written for the D1's CAN interface (NI-TLN-CONTROLS-DCDCCAN-R1). The D2 manual (Rev D) doesn't list its CAN messages. Stages 1–4 only listen, so they are safe either way. Don't send the D2 commands (stage 5) until Nomadic has confirmed that it obeys EMS_DCDC_CTRL (0x018) the same way as the D1.

What protects you

- The D2 itself: it stops on 12 V side over 16–18 V or under 8 V, 48 V side over 58–60 V or under 24 V, over 213 A, a short, or a board temperature over 120 °C (D2 manual)
- Your BMS and fuses on HV+ and LV+
- The driver's limits in config.ini, and the test timer
- **Your emergency stop: the fuse or breaker on a power lead.** The D2 has no KL15 to switch off. Opening the breaker on its LV+ or HV+ lead stops conversion immediately, whatever the Cerbo is doing

Before you start

- Fuses on HV+ and LV+ (D2 manual)
- A breaker or fuse on LV+ or HV+ you can reach quickly while testing
- A multimeter, plus a DC clamp meter if you have one
- Know your batteries' maximum charge voltages (12 V chassis and 48 V)
- The Cerbo's battery monitor in view (48 V current)
- The values on your D2's label: Nomadic can configure each unit differently

Stage	What	Risk
1	Simulation on the Cerbo	None: no converter, nothing sent on CAN
2	Wiring check, everything off	None: no power
3	Listen only	None: the Cerbo sends nothing
4	Installed in Automatic	None: the Cerbo sends nothing, and the D2 works as before
5	First commands with guard rails (after Nomadic confirms the CAN interface)	Low: small limits, 5-minute timer, hand on the breaker
6	Normal use	Only after stage 5 passes

1 Simulation: no converter at all

In Cerbo Setup, choose **9**. The real driver runs on your Cerbo against a pretend D2, and no CAN socket is ever opened. A real installation keeps running beside it.

Check	OK
"DC-DC D2 (SIMULATED)" appears in the Cerbo's switch pane, and in VRM's Switches	
Its web page opens at **http://<Cerbo IP>:8089**	
In Automatic the pretend 12 V side slowly drops. Below 12.5 V, Status changes to Autonomous and it charges back to about 13.4 V	
Buck: Status changes to Buck and the 12 V side climbs to your Buck voltage (13.5 V). Off: the current goes to zero	
Boost: the 48 V side rises a little, the 12 V side drops, and boost stops at 12.3 V	
Change the set-points and limits: values outside config.ini are refused	

Stop with Ctrl+C. The simulated device disappears. The pretend D2 follows the manual's logic, but it can't tell you how the real one reacts to commands. That's what stage 5 is for.

2 Wiring check, everything off

Check	OK
Fuses or breakers open on **both** LV+ and HV+ (the D2 powers itself from either side)	
Signal plug: only pins 1 (CAN_H) and 2 (CAN_L) used; pins 3–6 empty (no old D1 KL15/KL30 wires)	
Continuity: RJ45 pin 7 to white/brown, pin 8 to brown, pin 3 to white/green (guide step 2b)	
No continuity (no beep) between any of the three network-cable wires and LV+ or HV+	
CAN ground (white/green) to the D2 GND terminal: beeps	
RJ45 not in the Cerbo: white/brown to brown reads about 120 Ω (your resistor at the D2)	
RJ45 in the Cerbo, with the VE.Can terminator in the other socket: about 60 Ω	

3 Listen only

Power the D2 as normal (fuses or breakers on LV+ and HV+ closed). In Cerbo Setup choose **2** (find), then **3** (watch). The Cerbo sends nothing.

Check	What you saw	OK
Converter found, at the right speed (if it reports bus errors, switch speed as the guide says). If nothing is heard at all, ask Nomadic for the D2's CAN details before going on		
12 V side reading within about 0.2 V of your meter on the D2's LV+ to GND		
48 V side reading within about 0.5 V of your meter on HV+ to GND		
No alarms, or only alarms that make sense (for example low voltage when the battery really is low)		

4 Installed in Automatic

Choose **4** (install). The driver starts in Automatic and still sends nothing. Leave it running for a day.

Check	OK
Cards on the Cerbo, in VRM and on the web page, with sensible readings	
The D2 behaves exactly as before: buck when the 12 V side is low, standby otherwise	

5 First commands, with guard rails

Only once Nomadic has confirmed the D2 accepts EMS_DCDC_CTRL. Prepare: in Cerbo Setup choose **1**, set these test values, save, then choose **4**:

config.ini line	Test value	Why
`test_timer`	5	Buck/Boost switch to Off after 5 minutes
`buck_current_limit_max`	20	Small current while testing
`boost_current_limit_max`	10	Small current while testing
`boost_voltage`	Your 48 V battery voltage now + 0.5 V	Boost can only nudge it
`boost_voltage_max`	Your 48 V battery voltage now + 1 V	Nobody can set more

Conditions: 48 V side between 41 and 58 V; 12 V batteries healthy; someone at the LV+ or HV+ breaker; a meter on the D2's 12 V terminals. Do each step, watch for 1–2 minutes, and fill in the table.

#	Choose	Expect	What you saw	OK
5a	Off	Status: Standby/Idle. 12 V current about 0		
5b	Automatic	Status: back to the D2's own behaviour within a few seconds		
5c	Buck (13.5 V, ≤ 20 A)	Status: Buck. 12 V current ≤ 20 A (check with the clamp meter, and note which way is +). The 12 V side rises toward 13.5 V and never above about 13.7 V		
5d	Automatic	Back to the D2's own behaviour		
5e	Boost (now + 0.5 V, ≤ 10 A)	Status: Boost. 48 V current ≤ 10 A. The 48 V side rises a little and stops at the target. The 12 V side drops (it's the source). After 5 minutes the test timer switches to Off		
5f	Automatic	Back to the D2's own behaviour		
5g	Buck, then unplug the CAN cable	The hand-back test (guide Step 6): does the D2 keep going, stop, or return to its own logic? Plug it back in, then choose Automatic		

Stop at once (open the LV+ or HV+ breaker) if: any current goes above the limit you set; any voltage goes above your batteries' limits; a red Fault appears; the board temperature climbs fast (above about 80 °C); or anything smells, gets hot or sounds wrong. Then switch the Cerbo to Automatic before you close the breaker again. If the D2 then seems stuck on the last command, open the breakers on **both** LV+ and HV+ for a few seconds to restart its controller.

6 Normal use

Check	OK
Set `test_timer = 0`, and set the limits to what your wiring, fuses, batteries and the D2's label allow (option 1, then 4)	
Put `boost_voltage` back to your 48 V battery's charge voltage	
Use Automatic for everyday running, at least until you know how the D2 handles losing CAN (step 5g)	
Send Nomadic what you saw in 5g, and which way the current sign went in 5c	

Tested and signed off by: _____ Date: _____